

Herald



Herald — Issues

Field	Value
Revision	39
Created	2026-05-20
Last modified	2026-05-29
Status	active
Status summary	r39 (§11.4.19 atomic migration: HRD-090 dead-letter handling Completed→Fixed.md — all four closure layers met: code + unit + real-PG integration + §11.4.85 stress/chaos + paired §1.1 mutation gate (4/0) + e2e E138; HRD-147 filed for the governance-plane subscriber follow-up). Issues count steady at 32 (–HRD-090 +HRD-147). Prior r38 (HRD-146 filed — pherald listen dispatch hardening, from the 2026-05-29 "worker OOM" Phase-1 investigation that found the OOM premise did NOT match committed reality; the real latent risk is unbounded per-message dispatch growth — docs/research/oom-investigation-2026-05-29.md).** Prior r37 (HRD-090 dead-letter handling — code + unit + real-PG integration landed; stays in_progress pending §11.4.85 stress+chaos + paired §1.1 mutation gate + e2e invariant).** MoveToDeadLetter implemented (migration 000014_dead_letter_tasks, tx-atomic JSONB-snapshot move + terminal-mark, new 13th operational event class ClassQueueDeadLetter/QueueEvent/DadLetter() wired nil-tolerantly into pgxTaskRepository); 3 hermetic unit tests + TestRepoMoveToDeadLetter_RoundTripAndEmit real-PG integration (3 independent sinks: source terminal-state + dlq snapshot + real-bus event delivery) - race green; evidence docs/qa/HRD-090-20260529T052228Z/. NOT migrated to Fixed — closure bar (CONST-048 four-layer) not yet met. Prior r36 (§11.4.90 Obsolete graduation of 3 legacy NON-COMPLIANT live tests + Wave 8 Track B closure recorded against Fixed.md).** Three legacy live tests that fundamentally required manual operator action during execution graduate to Obsolete status per Universal §11.4.90 + Herald §108.e, with full Obsolete-Details lines (Since / Reason=superseded-by-later-mandate §11.4.98 / Superseding-item + triple-check evidence per §11.4.6 no-guessing): HRD-143 supersedes TestSubscribe_LiveBotAPI via HRD-140; HRD-144 supersedes tests/test_wave6_live_loop.sh via HRD-141; HRD-145 supersedes qaherald lifecycle --manual via HRD-142. Each Obsolete row cites a literal source-comment quote / file size / Cobra-flag line confirming the supersession is grounded in code that still exists, never a bare assertion. v0.6.0 tag (844f340) already covers HRD-139..142 in Fixed.md r17. Prior r35 (Tgram-completeness Wave C — HRD-137

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	§11.4.85 stress+chaos COMPLETE — SKIP→PASS conversion + atomic Issues→Fixed migration of all 6 tgram-completeness HRDs per §11.4.19).** Wave C closes the 6-HRD tgram-completeness wave (HRD-133..138). TestTgram_Stress_MultiRecipientFanOut already PASS (Wave-B-era multi-recipient fan-out stress). TestTgram_Chaos_GetUpdatesPollerResilience converted from t.Skipf SKIP-with-reason to real httpstest-driven fault injection via a one-line subscribe.go seam fix (threads a.baseURL into telebot.Settings.URL before NewBot — parallel to the ensureBot pattern at tgram.go:60-67; production path unchanged when a.baseURL == ""). 8-phase fault matrix (3× 500 + 2× 1.5s hangs honoring ctx-cancel + 1× mid-body Hijacker connection-close + 2× success), bot-username-aware self-filter bypass (success messages come from @someuser, not chaotestbot), assertion thresholds chosen deterministic-first: recoveryFloor=3 (observed recovery_event_count=6 every iteration), goroslack=16, settleWaitMs=2000. Deterministic across -race -count=3 per §11.4.50 — all 3 iterations IDENTICAL: recovery=6 fault(500=12 hang=8 close=3 ok=6) goroutines 3→4 (leaked=1) duration=12.07s. Full tgram suite 26/26 PASS, 0 SKIP post-fix (was 24/24 with chaos SKIPped). Evidence docs/qa/HRD-137-20260528T150000Z/stress_chaos/ (anchor preserved verbatim, §2 SKIP→PASS section appended, chaos_assertion.txt rewritten with measured PASS-state values). §11.4.92 5-pass: P1 captured-evidence per-iteration table; P2 blast-radius — subscribe.go baseURL seam is read-only when unused (production unaffected); P3 cross-feature — the seam mirrors existing ensureBot pattern, no novel state; P4 deep-research — telebot.v3 LongPoller retry behavior under fault (poller logs + continues, no internal retry); P5 anti-bluff — handlerHits, fault-counter atomics, ctx-aware subscribe return, no goroutine leak (delta=1) are all positive captured invariants from REAL Subscribe poller, not synthetic. Atomic migration: HRD-133, HRD-134, HRD-135, HRD-136, HRD-137 (this), HRD-138 ALL move Issues→Fixed in the same commit per §11.4.19 (Wave A/B previously parked them in_progress in Issues.md awaiting Wave C closure — now landed together). Prior r34 (Tgram-completeness Wave B combined — HRD-133 token-redaction + HRD-134 429 retry_after) marks BOTH Open→in_progress; Wave B is now CLOSED (5 of 6 tgram-completeness tasks done). Both HRDs land in ONE commit because they share commons_messaging/channels/tgram/send.go's error-and-retry paths atomically — splitting them would require partial-file commits with messy git add -p (precedent: C4+C5 / D-stragglers were the same shape). HRD-133 ships sanitizeTgramError (new errors.go) wrapping 6 error-return sites in Send + SendReply so the Bot API token NEVER appears in any error / log message; TestTgram_Send_ErrorDoesNotLeakToken plants 123456:supersecret-must-not-leak-9999 + asserts absence (mirrors TestSlack_Send_ErrorDoesNotLeakToken). HRD-134 ships withRetry429 wrapping all 3 telebot bot.Send call sites; on 429 + retry_after=N sleeps N seconds (ctx.Done()-aware), retries up to 3 attempts, clamps per-retry to 60s; retry_after extracted via errors.As(err, &telebot.FloodError{}). RetryAfter (telebot's typed error — no body parse). Composition: the retry's log lines + final error are themselves sanitized so 429 logging never leaks the token. Injectable sleeper field on Adapter for hermetic tests. 7 new tests -race -count=1 (2 HRD-133 + 5 HRD-134) → tgram suite now 24/24 PASS. Parts 4+5 of 6 in the tgram-completeness wave. Next: Wave C HRD-137 (multi-recipient fan-out stress + getUpdates resilience) on a now-stable send.go. Prior r33 (Tgram-completeness Wave A SA-c — HRD-138 operator deep-dive guide) marked HRD-138 Open→in_progress and closed Wave A (3 of 6 tgram-

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completeness tasks). docs/guides/TELEGRAM.md (810 lines, 11 §-sections, 58 subsections + 3 siblings .html/.pdf/.docx) is the per-module nano-detail operator guide called for by the operator memory feedback_nano_detail_docs. Sections: §1 architecture; §2 BotFather walkthrough; §3 env table + 3 chat_id discovery methods; §4 pre-deploy smoke-test commands; §5 attachment size limits + content-addressed inbox; §6 chat-type matrix (private/group/supergroup/channel id formats + bot permissions); §7 reply threading + attachment-fan-out rationale; §8 self-filter + echo-loop hazard (Wave 6 T12 M1); §9 12-entry troubleshooting cookbook (401/400/429/413/403/echo-loop/long-poll/409/privacy/webhook/zero-byte/wrong-inbox); §10 6-subsection pre-deploy audit (tick-boxes); §11 references (spec V3 + 13 HRDs + source paths). Cross-linked from the pre-existing docs/guides/messengers/TELEGRAM.md (HRD-011 setup, preserved untouched). Part 3 of 6 in the tgram-completeness wave — Wave A complete (HRD-135 + HRD-136 + HRD-138). **Next: Wave B HRD-133 + HRD-134** (combined subagent — both touch send.go: token-redaction symmetry + 429 retry_after honored), then **Wave C HRD-137** (multi-recipient stress+chaos). **Prior r32 (Tgram-completeness Wave A SA-b — HRD-136 webhook validator skeleton) marked HRD-136 Open→in_progress.** New commons_messaging/channels/tgram/webhook.go lands the Bot API webhook secret_token validation layer for the future V1.x webhook-path operator (spec V3 §11.1 alternative to getUpdates). WebhookHandler validates the X-Telegram-Bot-API-Secret-Token header via crypto/subtle.ConstantTimeCompare (timing-attack-safe), refuses empty secret at construction (NewWebhookHandler("",...) errors — §107 footgun guard), returns 401 with EMPTY body on header mismatch/missing (no token-existence-oracle leakage to probes), 400 on malformed JSON, 405 on wrong method, 500 (no body) on InboundHandler error. Decoded tele.Update is forwarded as the SAME InboundEvent shape OnText/OnEdited produce — unified downstream contract. **Skeleton-only:** wiring into pherald listen --webhook + setWebhook(secret_token=...) registration + TLS-bound public listener deferred to V1.x; the cryptographic layer is here + tested so the future flip is one-flag. 7/7 hermetic httptest tests -race: ValidSecret_Dispatches, InvalidSecret_Returns401_NoTokenLeak, MissingSecret_Returns401, GET_Returns405, MalformedJSON_Returns400, EmptySecretAtConstruction_ErrorsLoud, ConstantTimeCompare. Part 2 of 6 in the tgram-completeness wave. **Prior r31 (Tgram-completeness Wave A SA-a — HRD-135 lifecycle handlers) marked HRD-135 Open→in_progress.** commons_messaging/channels/tgram/subscribe.go gains 3 new event handlers per the operator's tgram-completeness mandate (Section 2 of the 2026-05-28 live-gap analysis): **OnEdited** rebuilds the same InboundEvent shape OnText does + Raw["edited"]=true; **OnMyChatMember** detects bot kicked/left/restricted status transitions (with Alice-left and promotion-to-Admin false-positive guards) + emits a grep-able log.Printf carrying ErrBotKicked text + the chat_id; **15 service-message endpoints** (UsersJoined/UserLeft/NewGroupTitle/etc.) registered to a single serviceHandler that emits nothing to InboundHandler but logs tgram: ignored service message kind=... chat_id=.... Sentinel var ErrBotKicked package-level so callers can errors.Is. 17/17 PASS -race -count=1 (3 new in subscribe_lifecycle_test.go + 14 existing, zero regression). Part 1 of 6 in the tgram-completeness wave. **Prior r30 (v0.6.0 Wave 7 T12 — operator guide; Wave 7 CODE-COMPLETE) marked HRD-121 Open→in_progress.** docs/guides/MESSENGER_CHANNELS.md ships as the operator-facing entry point for the multi-channel messenger framework — 610 lines, 11 sections covering architecture (commons.Channel outbound vs commons_messaging/channels.Channel inbound; ASCII fan-in diagram), HERALD_CHANNELS selector (fail-loud on unknown), Telegram + Slack live-setup walkthroughs (BotFather +

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api.slack.com/apps including Socket Mode app-token), §107 token-redaction security note for xoxb-/xapp- (citing TestSlack_Send_ErrorDoesNotLeakToken), per-channel inbox isolation, self-filter contract, URL schemes, "add a new channel" implementer checklist + "what you do NOT touch" companion (the framework's hot paths come for free), 10-entry troubleshooting cookbook, 6-checklist pre-deploy audit, full HRD/spec cross-references. Generated 3 siblings (.html/.pdf/.docx) via `REGEN_ALL=1 scripts/export_docs.sh`. **Wave 7 (T1-T12) is now code-complete:** HRD-110..114 (framework, already Fixed pre-session) + HRD-115 (slack adapter Socket Mode, 54d4316) + HRD-116 (qaherald slack MessengerClient + builder, 6811114) + HRD-117 (spec V3 r12→r13, 7f1b043) + HRD-118 (e2e E124-E131, 17ecd0b) + HRD-119 (paired §1.1 mutation gate 4/0, 6737040) + HRD-120 (multi-channel §11.4.85 stress+chaos 12/12, 86dfa1e) + HRD-121 (this row). Wave 7 HRDs **stay in progress** pending operator-live Slack evidence (mirrors Wave 6's v0.4.0 deferred-to-T13b pattern); v0.6.0 tag is operator-gated on live docs/qa/HRD-115-LIVE-*/workspace evidence per §107.x. **Prior r29 (v0.6.0 Wave 7 T11 — multi-channel §11.4.85 stress+chaos) marked HRD-120 Open→in progress.** Extended pherald/cmd/pherald/listen_stress_chaos_test.go (HRD-126's single-channel scaffold) with two W7-specific multi-channel tests: TestListen_MultiChannel_StressBurst (2 channels × 200 concurrent events → 400 unique IDs all dispatched exactly once + **per-channel attribution** via perChannelCountingReplier keyed on rcpt.Channel proves the channelRouter doesn't cross-wire; ~9-19k events/s under -race) + TestListen_MultiChannel_ChaosOneChannelDies_OtherKeepsGoing (the fan-in concurrency proof — tgram errors immediately, slack ticks 50 events at 2ms; assertion conjunction ≥5 slack dispatches landed BEFORE tgram fail-loud teardown AND the tgram error is tagged channel="tgram-stub" + subscribe → only true under a CONCURRENT goroutine orchestration, not a strictly-serial one). **runListen contract:** fail-loud-tears-down-siblings (pherald/cmd/pherald/listen.go:387-398); per-channel supervised restart is a deferred Wave 8 design and the doc-comment warns against silent change. 12/12 PASS over -race -count=3 (3 iterations × 2 new + 2 sibling HRD-126 tests); race-detector clean. Evidence anchors multichannel_no_drop_under_burst=1 (stress) + multichannel_concurrency_proven=1 (chaos) at docs/qa/HRD-120-20260528T110000Z/. **Prior r28 (v0.6.0 Wave 7 T10 — paired §1.1 mutation gate) marked HRD-119 Open→in progress.** tests/test_wave7_mutation_meta.sh ships 3 paired hermetic mutations against the Wave 7 multi-channel framework — **M1** blank IsSelfEcho (echo-loop hazard), **M2** drop slack channels.Register in init() (registry-resolution broken), **M3** drop slack thread_ts append (reply threading degrades) — each mutated build observed FAIL on the matching e2e detector + restored byte-for-byte + post-restore PASS. New TestSlackRegisteredViaInit detector added to slack_test.go (it asserts channels.Names() contains "slack" — the only test that would FAIL under M2 specifically; without it M2 would silently "no tests to run" PASS-bluff). Gate run = **4 PASS / 0 FAIL** (3 mutations + 1 quiescence assertion, no MUTATED W7 - residue in 3 production target files post-restore). Reuses Wave 6 canonical lockfile + trap + assert_restored + pre-flight guard verbatim. **FOREGROUND-ONLY** (§107.y / feedback_no_concurrent_mutation_gates operator mandate). **Prior r27 (v0.6.0 Wave 7 T9 — e2e multi-channel invariants) marked HRD-118 Open→in progress.** scripts/e2e_bluff_hunt.sh gains E124-E131 (8 invariants anchored to live test names already shipped in HRD-110..117): E124 channels registry (resolve + unknown-errors), E125 tgram self-filter post-Wave-7 generalization (TestSubscribeBotSelfFilter), E126 slack adapter (TestSlackSatisfiesChannel + auth.test + chat.postMessage + registry wiring +

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	sc_anchor to T6 docs/qa), E127 LIVE Slack triple-guarded (HERALD_SLACK_BOT_TOKEN + HERALD_SLACK_CHANNEL_ID + grep TestSlack_Live_Send — the third guard prevents the §107 "no tests to run = exit 0 = PASS-bluff" gotcha when env is set but no live-test exists), E128 per-channel inbox isolation + content-addressed idempotency, E129 generalized IsSelfEcho (Username + UserID + empty-self echo-loop hazard), E130 multi-channel pherald listen wiring (TestListenWiresHandlerAndExitsOnCancel), E131 qaherald messenger.Build dispatch (tgram + slack + unknown-errors-loud + sc_anchor to T7 docs/qa). Prior r26 (v0.6.0 Wave 7 T8 — spec V3 r12→r13) marked HRD-117 Open→in progress. Spec edits document the already-shipped Wave 7 framework + Slack: §11.0 gains the inbound-runtime extension note (commons_messaging/channels.Channel embeds commons.Channel + SendReplyGeneric + BotSelfIdentity + DownloadAttachment + init()-registered registry + generalized IsSelfEcho via `SelfIdentity{Kind: IdentityUsername
Issues	HRD-008, HRD-015, HRD-019..HRD-025, HRD-029..HRD-056, HRD-081, HRD-085..HRD-089, HRD-115..HRD-121, HRD-131, HRD-146, HRD-147 32 open / in-progress workable items. r35 (Tgram-completeness wave COMPLETE) atomically migrated HRD-133/134/135/136/137/138 Issues→Fixed per §11.4.19 — see Fixed.md. r34..r17 (v1.0.0 Batch A): HRD-018 (commons_constitution package — Evaluator+emit+migrations+bundle-hash+mode-ladder, completing the constitution_audit write-through that was the remaining §107 PASS-bluff: RunOutcome.Audited set true with nothing persisted), HRD-026 (bundle-hash now persisted on every emitted event AND audit row for replay), HRD-027 (mode-ladder runtime config + 3 admin REST endpoints GET/PUT /v1/compliance/modes[:rule_id]) closed atomically Task→done, migrated Issues→Fixed per §11.4.19. Evidence docs/qa/HRD-018-20260527T142556Z/ (real-PG emit→persist + admin-REST flip round-trips + §11.4.85 stress 512 emit→persist no-lost-writes + chaos 960 concurrent flips). Prior r16: HRD-122..HRD-130 (GAP-3 §11.4.85 stress+chaos suite) closed atomically as Completed + HRD-132 (Runner idempotency-window bug) closed as Fixed — 11 rows migrated Issues→Fixed; HRD-131 (SQLite SSoT migration Phase 1) remains OPEN (Phases 2-6 follow-on). r15: HRD-100 closed atomically (Wave 6 pherald inbound runtime — code-doc closure; live evidence operator-pending → T10b). r14: substrate upgrade only (Wave 4b TOON; no HRDs migrated). r12: HRD-016 closed atomically (Wave 3b pherald Runner live + e2e E37-E42 PASS or honest SKIP-with-reason when PG :24100 absent). r11: HRD-011 (Telegram live) closed atomically with live message_id=5 evidence. r10 doc-cleanup renumbered HRD-093 → HRD-099. r9: HRD-028 + HRD-098 closed (Wave 3a). Carry-over to Wave 3c: HRD-024 (iherald paging), HRD-033 (destructive-guard body), HRD-018..027 flavor-specific bindings, plus the rest of the §43 command catalogue.
Issues summary	HRD-001..HRD-007, HRD-009, HRD-009b, HRD-010, HRD-011, HRD-012, HRD-013, HRD-014, HRD-016, HRD-017, HRD-018, HRD-026, HRD-027, HRD-028, HRD-080, HRD-092, HRD-093, HRD-094, HRD-095, HRD-096, HRD-097, HRD-098, HRD-099, HRD-100, HRD-110, HRD-111, HRD-112, HRD-113, HRD-114, HRD-122, HRD-123, HRD-124, HRD-125, HRD-126, HRD-127, HRD-128, HRD-129, HRD-130, HRD-132, HRD-133, HRD-134, HRD-135, HRD-136, HRD-137, HRD-138
Fixed	
Fixed summary	see Fixed.md.
Continuation	see CONTINUATION.md.

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Open

Per Universal §11.4.74 every new row carries a Catalogue-Check line in its References cell (one of `reuse|extend|no-match <org/repo>@<sha>`). For HRDs whose catalogue-check is still pending, the row records `no-match (2026-05-20)` — operators reviewing the implementation PR will re-verify.

ID	Type	Status	Criticality	Title
HRD-143	task	Obsolete	low	TestSubscribe_LiveBotAPI legacy live test — superseded by Obsolete-Details: Since 2026-05-28; Reason=superseded-by-later-ma item=Universal §11.4.98 (full-automation anti-bluff mandate) + Herald (TestMTProto_Subscribe_AutonomousRoundTrip, Fixed.md r 367d58b); Triple-check evidence: (a) git-grep verified the test still ex commons_messaging/channels/tgram/subscribe_inte with the literal source comment "requires the operator to hand-send a r chat within the 60s window" (direct §11.4.98 violation — runtime hun test execution), (b) git-log shows 367d58b (HRD-140) introduced the autonomous replacement, (c) runtime path: the new HRD-140 test PAS end with zero human action per docs/qa/HRD-LIVE-MTPROTO-2 E135_subscribe/transcript.txt. Action: file is retained as (commented t.Skipf (§11.4.98 supersession) body) — op HRD-140's MTProto path.
HRD-144	task	Obsolete	low	tests/test_wave6_live_loop.sh legacy live closed-loop scr §11.4.98 + HRD-141. Obsolete-Details: Since 2026-05-28; Reason=s mandate; Superseding-item=Universal §11.4.98 + Herald §108.m + HI (TestMTProto_Wave6_AutonomousClosedLoop, Fixed.md r 7e50ed0+d7d6524+0797e85); Triple-check evidence: (a) ls te test_wave6_live_loop.sh confirms file exists (6782 bytes), (b (HRD-141) introduced the MTProto-driven autonomous closed-loop r based §11.4.98 autonomy assertion, (c) runtime path: HRD-141 PASSI bidirectional tgram.message in+cc.dispatch+cc.reply qa/HRD-LIVE-MTPROTO-20260528T125321Z/E136_wave6 transcript*.txt. Action: shell script retained as historical refere through HRD-141.
HRD-145	task	Obsolete	low	qaherald lifecycle --manual interactive mode — supersed Obsolete-Details: Since 2026-05-28; Reason=superseded-by-later-ma item=Universal §11.4.98 + Herald §108.m + HRD-142 (TestMTProto_Wave65_LifecycleAutonomous, Fixed.md r Triple-check evidence: (a) grep -nE '"--manual"' qaheral lifecycle.go:174 confirms cmd.Flags().BoolVar(&f.M false, ...) is wired in the Cobra subcommand, (b) git-log shows introduced the 3-scenario MTProto-driven autonomous replacement (F paths, journal-based assertions), (c) runtime path: HRD-142 PASSED subtests PASS per docs/qa/HRD-LIVE-MTPROTO-20260528T

ID	Type	Status	Criticality	Title
				E137_wave65_lifecycle/transcript*.txt. Action: --ma hoc operator debugging but explicitly NOT a §11.4.98-compliant test p MTProto build-tag path from HRD-142.
HRD-015	task	open	low	Inheritance gate I8 invariants for Go scaffold (go.work + commons/typ passes) cherald constitution bindings — cherald/internal/bindin cherald-owned §42.3 rules as concrete commons_constitution. through 5 event classes (.policy.violation/.gate.failed/.gate.recovered/. over the Batch-A emit+persist+audit foundation; POST /v1/compl surface drives detect→emit→persist→audit, visible on the existing GE pull surface. Vertical-slice bespoke detectors landed (§11.4.29 naming §11.4.44 doc-revision-header, §11.4.18 script-docs, §11.4.73 spec-drift remaining 25 rules use the shared marker-check mechanism (verdict d catalogue command / sweep) with the full emit-routing framework in p remaining bespoke detector is per-rule follow-up.
HRD-019	task	in_progress	high	sherald host-safety + repo-safety constitution bindings — sher al bindings Pipeline registers the 15 sherald-owned §42.3 host/repo-sa §11.4.24/.26/.32/.36/.41/.47/.71, §12.1/.2/.3/.6) as concrete commons_constitution.Evaluator impls routed through 6 S (.host.safety.breach/.repo.safety.breach/.gate.re over the Batch-A emit+persist+audit foundation, mirroring the proven pattern. The 3 named DETECTION hooks landed as REAL pure class detector (§9.1: rm/git-reset/clean without backup → .repo.safety interceptor (§11.4.41/§9.2: force-push without merge-first OR session- .repo.safety.breach), mem-budget watcher (§12.6: reported u .host.safety.breach) — plus forbidden-host-op (§12.1) + prer remaining rows. CRITICAL §12-safety: every detector DETECTS+RE string description; NONE performs the destructive op, force-pushes, s allocates memory (the mem-budget watcher reads a fabricated fraction §43 command bodies (HRD-033/046/056) supply the live op-intercept
HRD-020	task	in_progress	high	bherald CI/test bindings — bherald/internal/bindings P bherald-owned §42.3 CI/test gate-result rules (§1, §1.1, §11.4.2/.3/.4/.5/.7/.9/.13/.14/.24/.27/.30/.39/.43/.46/.48–.52/.67) as con commons_constitution.Evaluator impls routed through 3 e (.gate.failed/.gate.recovered via the build/test gate-result .policy.violation+ .repo.safety.breach for the shared row) over the Batch-A emit+persist+audit foundation, mirroring the pr Pipeline pattern. bherald is the BUILD/CI flavor; it emits CI/test gate- that integrate with the consuming project's CI workflow. 3 bespoke pu result classification (§1/§11.4.50 — PASS/FAIL/flaky from a recorded .gate.failed/.gate.recovered), test-tier-verify (§11.4.27/§4 matrix; a missing tier FAILs), anti-bluff-PASS detection (§11.4.2/.5 — WITHOUT a captured-evidence artefact is a §11.4 PASS-bluff → FAI shared marker-check mechanism (verdict decided upstream by the §43 full emit-routing framework in place.
HRD-021	task	in_progress	middle	rherald release constitution bindings — rherald/internal/b registers the 4 rherald-owned §42.3 release-lifecycle rules (§4 tag mirr multi-format export [shared with cherald], §11.4.38 installable-asset ex retest before release tag) as concrete commons_constitution.E

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HRD-023 task in_progress middle

HRD-024 task in_progress middle

HRD-025 task in_progress low

HRD-029 task in_progress middle

HRD-030 task in_progress middle

HRD-031 task in_progress middle

HRD-032 task in_progress low

HRD-033 task in_progress high

through 2 event classes (`.release.gate.blocked` via the existing `ReleaseGateBlocked` emitter surface for §4/§11.4.38/§11.4.40 + `.release.gate.blocked` for the shared §5 changelog row; a release recovery maps to the shared `release` companion since the release class family has no distinct "recovered" le `emit+persist+audit` foundation, mirroring the proven HRD-019/020/021 the RELEASE flavor. 4 bespoke PURE detectors landed: `tag-mirror-parent` but missing on any owned mirror, `with_tag<mirrors, → .release.gate.blocked`), `changelog-conformance` (§5 — non-`changelog` OR stale multi-format export → `.policy.violation`), (§11.4.40 CRITICAL — tag attempted without a green all-8-§40.2-tier `.release.gate.blocked`), `installable-asset-evidence` (§11.4.38 — install check → `.release.gate.blocked`). CRITICAL §12-safe CLASSIFIES a recorded release-op outcome string; NONE tags, push downloads, or installs — the live release-op interception is scope-lock (HRD-031/032/045).

`pherald` project bindings (§2 `commit+push`, §3 submodule propagation §11.4.11/.15/.21/.22/.34/.37/.42/.55/.66/.71/.74) — Batch B: `pherald bindings/ Pipeline + §42.3-owned rule set + 5 bespoke PURE detector discipline / submodule-propagation-order / pre-push-fetch-guard / fetch through the commons_constitution Evaluator+audit foundation; -race - Live op-interception deferred to §43 HRD-029/030/043/044/049/053.`

`iherald` constitution-rule escalation bindings (§11.4.10/.10.A `credential` + §11.4.66 operator-blocked escalation) — Batch B: `iherald/inter Pipeline routing .credential.leak` (forced-critical) + `.policy detectors` wired through commons_constitution Evaluator+audit; -race no-real-secret attestation. Live paging deferred to §43 / HRD-024-paging

`scherald` scheduled-audit bindings (§11.4.45 periodic Status.md sweep compliance digest) — Batch B: `scherald/internal/bindings .policy.violation + 3 bespoke detectors` (status-sweep / digest through commons_constitution Evaluator+audit; -race -count=3 green scheduler deferred to §43 HRD-047.

§2 `pherald commit-push` — single-entrypoint locked `commit + C: implemented as gitops command` (§11.4.74 wrap of git binary); --en hermetic temp-repo tests -race green

§3 `pherald submodule propagate` — owned-submodule wall Batch C: implemented as gitops command (§11.4.74 wrap of git binary binding; hermetic temp-repo tests -race green

§4 `rherald tag mirror` — assert tag exists on every owned sub implemented as `rherald §43 release command` (`rherald/cmd/rherald release_cmds.go`) feeding the HRD-022 `rherald` binding; reuses s hermetic -race green

§5 `rherald changelog generate` — Conventional Commits - <v>.md + multi-format export — Batch C: implemented as `rherald §43` (`rherald/cmd/rherald/release_cmds.go`) feeding the HRD-022 reuses shared commons/gitops; hermetic -race green

§9.1 `scherald destructive guard <op>` — wrap `rm/git-res prerequisite checks` — Batch C: implemented as `scherald §43 command` (`scherald/sysops_cmds.go`) feeding the HRD-020 `scherald` binding (detects+classifies+exits-nonzero, never performs the op); hermetic -ra fix so the CLI runs without a JWT secret

ID	Type	Status	Criticality	Title
HRD-034	task	in_progress	middle	§9.3 sherald backup-snapshot <target> — hardlinked snapshot straggler: implemented in sherald/cmd/sherald/sysops_cmds.go; refuses overwrite, classifies §9.3 SubjectBackup → .gate.recover; equality -race green
HRD-035	task	in_progress	middle	§11.4.2 bherald evidence capture <test_id> — captured Batch C: implemented as bherald §43 build/CI command (bherald/build_cmds.go) feeding the HRD-021 bherald binding; hermetic -race green
HRD-036	task	in_progress	high	§11.4.10 cherald creds scan — gitleaks/trufflehog integration, .credential.leak — Batch C: implemented as cherald §43 verify (cherald/cmd/cherald/checkops_cmds.go) feeding the HRD-019 cherald binding; detect-only; hermetic -race green (routed to §11.4.10 credential-leak; fix so the secret value printed)
HRD-037	task	in_progress	low	§11.4.12 cherald docs sync — regen Issues_Summary / Fixed_Status_Summary — Batch C: implemented as cherald §43 docs-pipeline cmd/cherald/docops_cmds.go feeding the HRD-019 cherald binding; --apply to regenerate; ships the §107 lazy-verifier fix so the CLI runs hermetic -race green
HRD-038	task	in_progress	low	§11.4.18 cherald script-docs check — assert sibling .md for Batch C: implemented as cherald §43 verify/check command (cherald/cmd/cherald/checkops_cmds.go) feeding the HRD-019 cherald binding; detect-only (routed to §11.4.60 — §11.4.62 has no cherald rule)
HRD-039	task	in_progress	low	§11.4.19/.23 cherald fixed align+cherald colorize — HTML colorizer — Batch C: implemented as cherald §43 docs-pipeline cmd/cherald/docops_cmds.go feeding the HRD-019 cherald binding; --apply to regenerate; ships the §107 lazy-verifier fix so the CLI runs hermetic -race green
HRD-040	task	in_progress	high	§11.4.26 sherald constitution pull — wrap fetch + rebase .bundle.updated — Batch C: implemented as sherald §43 command sherald/sysops_cmds.go feeding the HRD-020 sherald binding; (detects+classifies+exits-nonzero, never performs the op); hermetic -race green fix so the CLI runs without a JWT secret
HRD-041	task	in_progress	middle	§11.4.27 bherald test-tier verify — 8-tier matrix verification implemented as bherald §43 build/CI command (bherald/cmd/bherald/build_cmds.go) feeding the HRD-021 bherald binding; hermetic -race green
HRD-042	task	in_progress	low	§11.4.31 cherald submanifest verify — Submodule-Dependency Batch C: implemented as cherald §43 verify/check command (cherald/cmd/cherald/checkops_cmds.go) feeding the HRD-019 cherald binding; detect-only
HRD-043	task	in_progress	high	§11.4.36 pherald install-upstreams — install_upstreams wrapper implemented as gitops command (§11.4.74 FindScript-resolved constitution HRD-023 binding; hermetic temp-repo tests -race green
HRD-044	task	in_progress	middle	§11.4.37 pherald fetch-guard — pre-edit fetch + rebase enforcement implemented as gitops command (§11.4.74 wrap of git binary); --emit temp-repo tests -race green
HRD-045	task	in_progress	high	§11.4.40 rherald gate retest — pre-tag full-suite retest gate -rherald §43 release command (rherald/cmd/rherald/release_cmds.go) HRD-022 rherald binding; reuses shared commons/gitops; hermetic -race green
HRD-046	task	in_progress	high	§11.4.41 sherald force-push gate — merge-first + per-session Batch C: implemented as sherald §43 command (sherald/cmd/sherald/sysops_cmds.go) feeding the HRD-020 sherald binding; PRE-FLIGHT

ID	Type	Status	Criticality	Title
				(detects+classifies+exits-nonzero, never performs the op); hermetic -race fix so the CLI runs without a JWT secret
HRD-047	task	in_progress	middle	§11.4.45 / .56 <code>scherald status-digest</code> — periodic <code>Status.md</code> regen — Batch C straggler: implemented in <code>scherald/cmd/scherald/docops_cmds.go</code> feeding the HRD-025 <code>scherald</code> binding (§11.4.45 <code>status-sweep</code> ; detect-by-regenerates <code>Status_Summary</code>); hermetic -race green
HRD-048	task	in_progress	low	§11.4.53 <code>cherald fixed-summary sync</code> — standalone Fixed Summary Batch C: implemented as <code>cherald §43 docs-pipeline</code> command (<code>cherald/cmd/docops_cmds.go</code>) feeding the HRD-019 <code>cherald</code> binding; detect-by-regenerate; ships the §107 lazy-verifier fix so the CLI runs without a JWT secret green
HRD-049	task	in_progress	middle	§11.4.55 <code>pherald reopen <HRD-NNN></code> — Issues→Fixed reverses Batch C: implemented as <code>gitops</code> command (the Wave 6.5 inbound Fixed Summary emit feeds HRD-023 binding; hermetic temp-repo tests -race green
HRD-050	task	in_progress	middle	§11.4.59 <code>cherald readme sync</code> — README doc-link regen + regenerate Batch C: implemented as <code>cherald §43 docs-pipeline</code> command (<code>cherald/cmd/docops_cmds.go</code>) feeding the HRD-019 <code>cherald</code> binding; detect-by-regenerate; ships the §107 lazy-verifier fix so the CLI runs without a JWT secret green
HRD-051	task	in_progress	high	§11.4.60 <code>cherald composite-gate</code> — canonical implementation of COMPOSITE-SYNC — Batch C: implemented as <code>cherald §43 verify/check</code> command (<code>cherald/cmd/cherald/checkops_cmds.go</code>) feeding the HRD-019 <code>cherald</code> binding; detect-only; hermetic -race green
HRD-052	task	in_progress	middle	§11.4.65 <code>cherald export</code> — bulk Markdown export wrapper (md2html) implemented as <code>cherald §43 docs-pipeline</code> command (<code>cherald/cmd/docops_cmds.go</code>) feeding the HRD-019 <code>cherald</code> binding; detect-by-regenerate; ships the §107 lazy-verifier fix so the CLI runs without a JWT secret green
HRD-053	task	in_progress	high	§11.4.71 <code>pherald pre-push</code> — fetch + investigate + integrate hooks as <code>gitops</code> command (§11.4.74 wrap of git binary); --emit feeds HRD-023 binding; repo tests -race green
HRD-054	task	in_progress	middle	§11.4.73 <code>cherald spec-version check</code> — Revision-vs-edits check implemented as <code>cherald §43 verify/check</code> command (<code>cherald/cmd/cherald/checkops_cmds.go</code>) feeding the HRD-019 <code>cherald</code> binding; detect-only; hermetic -race green
HRD-055	task	in_progress	middle	§11.4.74 <code>cherald catalogue-check <pr></code> — scan PR for Catalogue runner over vasic-digital + HelixDevelopment — Batch C: implemented as <code>cherald §43 docs-pipeline</code> command (<code>cherald/cmd/docops_cmds.go</code>) feeding the HRD-019 <code>cherald</code> binding; detect-only; hermetic -race green
HRD-056	task	in_progress	high	§12.6 <code>sherald mem-budget watch</code> — daemon-mode 60% threshold <code>.host.safety_breach</code> — Batch C: implemented as <code>sherald §43 cmd/sherald/sysops_cmds.go</code> feeding the HRD-020 <code>sherald</code> binding; GATE (detects+classifies+exits-nonzero, never performs the op); hermetic -race fix so the CLI runs without a JWT secret
HRD-081	task	open	low	Extend <code>digital.vasic.containers/pkg/compose</code> to detect docker compose runtime and either emit runtime-appropriate flags (none for docker compose) or fall back to host-side healthcheck polling. Composes with <code>compose.WithWait(true)</code> works across both backends. Also fix <code>podman-compose ps</code> output (returns 0 services even when containers are running directly). Per §11.4.76: extend upstream submodule, never reimplement

ID	Type	Status	Criticality	Title
HRD-085	task	in_progress	middle	Implement single-task admin operations on <code>commons_infra.pgxTaskRepository</code> : <code>GetByID</code> , <code>Update</code> , <code>Delete</code> . Required for Requeue + MoveToDeadLetterQueue operator tooling. Currently all three return <code>ErrUnsupported</code> (HRD-085). Scope: BackgroundTask row by ID; full-row Update including JSONB column decision per §107.
HRD-086	task	in_progress	middle	Implement worker-side running-task state mutations on <code>commons_infra.pgxTaskRepository</code> — <code>UpdateStatus</code> , <code>UpdateHeartbeat</code> , <code>SaveCheckpoint</code> . Required by River worker reaper. Currently all four return <code>ErrUnsupported</code> (HRD-086). Scope: guarded by valid-state-machine check; progress as (<code>float64</code> , <code>messageID</code> , <code>updated_at</code> = <code>now()</code>) writeback; checkpoint as opaque []byte on task row.
HRD-087	task	in_progress	low	Implement admin / stats / audit reads on <code>commons_infra.pgxTaskRepository</code> : <code>GetByStatus</code> , <code>GetPendingTasks</code> , <code>CountByStatus</code> , <code>GetTaskHistory</code> , <code>v1/queue/*</code> introspection endpoints + <code>Peek</code> delegate. Currently all return <code>ErrUnsupported</code> (HRD-087). Scope: status-filtered paginated read; <code>Peek</code> ; status-bucketed counts; per-task execution-history rows from the <code>task_execution_history</code> table.
HRD-088	task	in_progress	middle	Implement queue-recovery reads on <code>commons_infra.pgxTaskRepository</code> : <code>GetStaleTasks</code> , <code>GetByWorkerID</code> . Drives the stuck-task reaper loop. Currently both return <code>ErrUnsupported</code> (HRD-088). Scope: guarded against <code>updated_at</code> ; per-worker assignment lookup for re-dispatch.
HRD-089	task	in_progress	low	Implement resource-monitor I/O on <code>commons_infra.pgxTaskRepository</code> : <code>SaveResourceSnapshot</code> , <code>GetResourceSnapshots</code> . Persists resource samples for capacity-planning + post-mortem. Currently both return <code>ErrUnsupported</code> (HRD-089). Scope: append-only insert keyed by (<code>task_id</code> , <code>capacity</code>); paginated read in reverse-chronological order. Schema target: a <code>task_resource</code> table not yet in the §9.6 migration set — adding it is part of this HRD.
HRD-147	task	open	low	Governance-plane subscriber for <code>.queue.dead_letter</code> — wire a <code>Consumer</code> that drains the HRD-090 dead-letter events into the <code>const t</code> (equivalent durable store) so dead-letter moves are auditable beyond the queue; emits the event + proves fan-out to a REAL bus subscriber in its integrity; adds the production subscriber + persistence and wires the emitter into <code>pgxTaskRepository</code> construction (boot.go currently uses the nil-emitter); the plane has no governance bus; this HRD decides where the bus + subscriber binding plane vs a dedicated consumer).
HRD-115	task	in_progress	high	Wave 7 T6 — Slack channel adapter (Socket Mode): <code>commons_messaging.slack/</code> implements <code>channels.Channel</code> — <code>New/NewWithBaseURL</code> (parsing <code>slack://<bot-token>@<workspace>/<channel-id></code>), <code>Send</code> (<code>chat.postMessage</code>), <code>SendReply</code> (<code>thread_ts</code>) + <code>SendReplyGetFiles</code> (<code>w/files.uploadV2</code> attachment fan-out), <code>Subscribe</code> (Socket Mode), <code>Unsubscribe</code> (auth.test → <code>IdentityUserID</code> , cached), <code>DownloadAttachment</code> (<code>files.getAttachment/inbox/slack/<sha>.<ext></code> via <code>channels.WriteContentAddressableFile</code>). Vendored submodules/slack-go @ v0.16.0 (commit 209ec09). Build: <code>commons_messaging/go.mod</code> replace per Herald §11.4.74. 22 headers (incl. subtests) -race green; inheritance gate 15/0 post-gitmodules-ec2. TestSlackSatisfiesChannel confirms interface satisfaction.
HRD-116	task	in_progress	middle	Wave 7 T7 — qaherald Slack MessengerClient + channel-keyed <code>internal/messenger/{slack.go, builder.go}</code> : raw-HTTP + 11 <code>MessengerClient</code> methods (<code>Me/Send/SendPhoto/SendDocument</code>

ID	Type	Status	Criticality	Title
				WaitForReply/Download/Preflight/GetChatMember/Close); GetUpdate conversations.history+oldest=<ts> watermark (Slack has MessageID = dotted-float ts string; channel-keyed Build (BuildC unknown→error). Token NEVER in errors — sanitizeError() c TestSlack_Send_ErrorDoesNotLeakToken pins it with a pl SUPERSECRET-must-not-leak-7777. 30 hermetic httptest test time pin var _ MessengerClient = (*SlackClient)(ni CLEANUP: ships files.upload V1 (deprecated by Slack 2025-Q — functionally equivalent for the httptest seam, swappable without A slack.go header.
HRD-117	task	in_progress	low	Wave 7 T8 — spec V3 r12→r13: §11.0 channels.Channel inbound §32.2 Telegram/Slack marked LIVE + §43 multi-channel pherald the Wave 7 framework + Slack adapter (T6) + qaherald Slack client (T catches up to shipped code (no new product code in T8).
HRD-118	task	in_progress	high	Wave 7 T9 — e2e_bluff_hunt.sh E124-E131 multi-channel invariants: resolution, E125 tgram self-filter post-Wave-7 (pure-refactor invariant (interface satisfied + auth.test + chat.postMessage wire bytes + registry Slack (env-gated HERALD_SLACK_BOT_TOKEN+HERALD_SLACK_CHANNEL_ID+ triple-guard to prevent silent "no tests to run" PASS-bluff), E128 per-c E129 generalized IsSelfEcho (Username+UserID+empty-self), E130 n wiring, E131 qaherald messenger.Build dispatch. sc_anchor on the T6-
HRD-119	task	in_progress	high	Wave 7 T10 — paired §1.1 mutation gate tests/test_wave7_mu PASS / 0 FAIL): M1 blank IsSelfEcho → TestIsSelfEchoUs hazard caught) → restored → PASS; M2 drop slack channels.Reg TestSlackRegisteredViaInit FAILs → restored → PASS; M append in send.go → TestSlackSendReplyUsesThreadTS + quiescence assertion (no MUTATED W7 - residue in 3 production ta Reuses Wave 6 canonical .git/MUTATION_IN_PROGRESS lockfil flight dirty-tree-and-residue guard + byte-for-byte assert_restore ONLY (§107.y / no-concurrent-mutation operator mandate).
HRD-120	task	in_progress	high	Wave 7 T11 — §11.4.85 multi-channel stress + chaos. Extended HRD- pherald/listen_stress_chaos_test.go (NOT duplicated TestListen_MultiChannel_StressBurst (2 channels × 200 ONE runListen — pins no-loss/no-double + per-channel attribution perChannelCountingReplier keyed on rcpt.Channel prov NOT cross-wire; throughput ~9-19k ev/s under -race; clean shutdown TestListen_MultiChannel_ChaosOneChannelDies_0th fires 1 event then errors; slack ticks 50 events at 2ms — pins concurr via conjunction: ≥5 slack dispatches landed BEFORE fail-loud teardov surfaces tagged channel="tgram-stub" + subscribe). 12/12 -count=3 (3 iterations × 2 new + 2 sibling HRD-126); race-detector found: fail-loud-tears-down-siblings (per-channel supervised restart i doc-comment warns against silent change). Evidence anchors: multichannel_no_drop_under_burst=1 (stress) + multichannel_concurrency_proven=1 + multichannel_one_channel_death_surfaced=1 (chaos).
HRD-121	task	in_progress	low	Wave 7 T12 — docs/guides/MESSENGER_CHANNELS.md oper sections + 3 siblings .html/.pdf/.docx): §1 architecture (commons.Char commons_messaging/channels.Channel inbound + registry + IsSelfEc ASCII fan-in diagram), §2 HERALD_CHANNELS selector (fail-loud or

ID

Type Status

Criticality Title

HRD-131 task open

middle

Telegram live setup (BotFather → token → chat-id), §4 Slack live setup (Socket Mode + scopes + xoxb-/xapp- tokens; §107 token-redaction TestSlack_Send_ErrorDoesNotLeakToken), §5 inbox isolation {HERALD_INBOX_DIR}/<channel>/<sha256>.<ext>, §6 an (cross-the-wire-then-cache contract), §7 URL schemes for both adapters "channel" implementer checklist + companion "what you do NOT touch come for free), §9 10-entry troubleshooting cookbook (echo-loop / chat invalid_auth / wrong inbox / HERALD_CHANNELS=unknown), §10 deploy audit mirroring OPERATOR_CREDENTIALS.md style, §11 re V3 §11/§32.2/§43 + Wave 7 plan + adapter source paths + every HRD REGEN_ALL=1 bash scripts/export_docs.sh (new-guide)
Migrate text HRD trackers to versioned SQLite single-source-of-truth (6-phase) — DB committed as authoritative SSoT per operator mandate (item): file the scope-locked tracking HRD. Phase 2: scaffold/adopt the already lives in the constitution submodule (constitution/scripts/items/) — Herald references it per §11.4.74 catalogue-first and never sync md→db (load docs/Issues.md / docs/Fixed.md / *_Summary.md / docs/CONTINUATION.md §3 into docs/.workable) sync db→md byte-identical (round-trip regeneration so MD↔DB drift impossible). Phase 5: generator shims (the *_Summary.md variants become derived from the DB). Phase 6: prohibit text-direct edits (release the only authoring surface). Operator divergence from the parent §11.4 regeneration default: for Herald the DB IS the authoritative artefact — docs/.workable_items.db is version-controlled (committed), but only the transient WAL/SHM sidecars (docs/.workable_items.db) are gitignored ("We should not git ignore our workable items database since the truth regarding items we have to do on the project!"). **Scope-lock: this row) + Phase 2 adoption ONLY.** Phases 3-6 (md→db sync, db→md sync, generator shims, text-direct-edit prohibition) are follow-on work-items completes, so the §106 spec-change / §107.x evidence invariants are not **2026-05-29 re-assessment:** Phase 2 remains EXTERNALLY BLOCKED. scripts/workable-items/ is still a non-functional scaffold (0% implemented, go build succeeds but every command exits 2 "not yet re-assessment docs/research/workable-items-phase2-re-assessment-2026-05-27.md). Herald cannot proceed beyond Phase 2 reimplementing the binary, which §11.4.74 forbids. The docs/.workable canonical docs/workable_items.db path reconciliation is DEFERRED. CLAUDE.md §11.4.95 ("reconcile to the canonical path when the DB is reconciling now would churn 4 governance docs (inheritance-gate risk exist and could create a new dot/no-dot split.

HRD-146 task open

middle

pherald listen dispatch hardening — bound per-message resource inbound→claude_code.Dispatch path. Latent unbounded-growth 2026-05-29 "worker OOM" Phase-1 investigation (docs/research/workable-items-investigation-2026-05-29.md): dispatch.go uses cmd.StdoutBuffer (no stdout buffering in memory) with NO per-message timeout beyond the subscribe.go runs handlers on telebot dispatch goroutines. Under heavy + sustained inbound load, goroutines + stdout buffers could accumulate. **Hypothesis-grade (NOT a confirmed defect)** — untested under real stress tests use stub dispatchers). Scope: add a per-message context to dispatch + cap/stream child stdout (cmd.Output()→bounded reader) chaos test driving a deliberately-slow fake claude to prove bounded

ID Type Status Criticality Title

NOTE: the originating "hourly OOM + telemetry-drop + 5-path triage
NOT match committed reality (no hourly worker, no OTel telemetry, n

In progress

ID	Type	Status	Criticality	Title	Opened	Last update	References
HRD-008	task	in_progress	middle	Operator- side quickstart compose validation (Postgres + Redis + OTel + pherald container)	2026-05-20	2026-05-20	spec V3 §26.5 — scaffold shipped, live end-to-end run pending operator; Catalogue-Check: no-match (2026-05-20) — bespoke to Herald

Blocked

(none)

Conventions

See [Fixed.md](#) for closed items + Universal §11.4.12/.15/.16/.19/.33/.55/.73/.74 composition
rules.